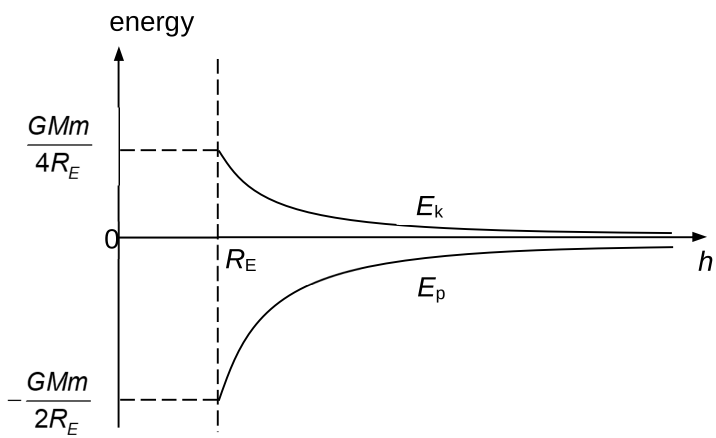
		answer
3(a)	The gravitational potential at a point is the work done per unit mass required in moving a small test mass from infinity to that point.	[2] or 0

3(b)(i)		[1] for E_p [1] for E_k -1 if y-axis not labelled
3(b)(ii)	The gravitational force on the satellite provides for the centripetal force required to move it in a circular path. $F_G = F_C$ $\frac{GMm}{R^2} = mR\omega^2$ $\frac{(6.67\times10^{-11})(6.0\times10^{24})}{R^3} = \left(\frac{2\pi}{12\times60\times60}\right)^2$ $R = 2.66\times10^7$ m $h = 2.66\times10^7 - 6.4\times10^6$ $= 2.02\times10^7$ m $\approx 2.0\times10^7$ m	[1] [1] for correct substitution for $F_G = F_C$ [1] for h
3(b)(iii)	In order for the satellite to orbit at a larger h and orbital radius, the <u>total energy required in the satellite is larger</u> as it is equals to $-\frac{GMm}{2r}$. Hence the work done required on the satellite is <u>positive</u>.	[1] [1]
4(a)(i)	The waves reaching at AB will have different path length. When	[1] for path